(PV-SRS-01) SwSRS

Document ID: `PV-SRS-01`
Product: `Portview`
Document Status: `Released`

Document Overview

This document defines the software requirements for Portview.

Document notes:

- The current structure follows the functional scope visible in the existing Portview software requirement set.
- Portview-only content is retained in this document set.

Document Approval

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Revision History

Rev.	Date	Description
0.0	2012.03.07	Initial version
0.1	2015.01.12	User interface update
0.2	2016.01.19	New function implemented
0.3	2017.01.13	System issue revision
0.4	2018.01.12	New device added
0.5	2019.01.21	Device compatibility update
0.6	2020.01.30	Device compatibility update
0.7	2020.10.08	GUI update
0.8	2021.02.26	Linkage usability improvement
0.9	2021.09.10	Program issue update
1.0	2021.10.08	Device compatibility update
1.1	2022.03.04	Connection-status improvement
1.2	2023.01.04	Increased image capacity
1.3	2023.04.21	Device compatibility update
1.4	2024.01.08	Device compatibility update
1.5	2024.01.10	Document number changed from 603 to Z01 according to OP-709
1.6	2025.07.24	Updated general requirements and added GUI translation requirement

1. Purpose

This document defines the software requirements for Portview and describes the software behavior needed to satisfy the intended product workflow.

2. References

The software requirement set is supported primarily by:

- `RS for Portview`
- `SwSDS for Portview`
- `SV-04` Software Verification Plan
- `SV-05` Software Verification Report

3. Definitions And Acronyms

Common terminology should be normalized in the controlled release set. Acronyms visible in the current requirement content include:

Acronym	Meaning
`DICOM`	Digital Imaging and Communications in Medicine
`GUI`	Graphical User Interface
`TWAIN`	Imaging-device interface standard used for import

4. General Requirements

4.1 Product Perspective

Portview is workstation software used to acquire, store, display, review, and exchange dental images.

The current software context includes:

- patient management functionality
- device-service connectivity checks
- image-viewer presentation and annotation tools
- export and communication services
- GUI translation support

The authored scope excludes ImageProcess-specific processing requirements.

4.2 Product Functions

The software supports:

- patient-oriented image acquisition and review
- image storage and retrieval
- diagnostic viewing and comparison
- annotation and measurement
- print, export, and DICOM communication
- device connectivity status and operational logging

4.3 Intended User Context

The software operator is expected to understand clinical or hospital workflow and to be familiar with the applicable user manual.

4.4 General Constraints

The software relies on:

- supported workstation operating system configuration
- supported acquisition devices and drivers
- controlled local storage and communication interfaces

The currently visible constraints are:

Constraint Area	Current Controlled Statement
Operating system	Windows XP Service Pack 2 (x86) or higher
Concurrent database users	Depends on the operating system configuration
Display resolution	Greater than `1024 x 768`
Database-server memory	Greater than `4 GB`

5. Functional Requirement Groups

5.1 User And Account Functions

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-001`	User authentication using Windows account	`SDS-001`	Application authenticates user through the Windows OS account; no separate login is required
`SRS-002`	Patient registration	`SDS-001`	New patient record is created with validated demographic fields and persisted to the database
`SRS-003`	Modify patient information	`SDS-001`	Modified patient fields are saved correctly; invalid modifications produce an error message
`SRS-004`	Search patient	`SDS-001`	Search by patient identifiers returns matching records; no-match condition is handled gracefully
`SRS-005`	Delete patient	`SDS-001`	Patient record is removed after confirmation; active-record conflicts are prevented

5.2 Acquisition Functions

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-006`	Acquire image from IO sensor	`SDS-024`	Sensor image is acquired and displayed under the selected patient without data corruption
`SRS-007`	Select the device that captures the image	`SDS-023`, `SDS-028`	Selected device becomes active and viewer state follows the selected device type
`SRS-008`	Display status of the device	`SDS-026`	Device status changes are reflected correctly between connected and disconnected states
`SRS-009`	Acquire image from a file	`SDS-002`	Selected file is imported and displayed correctly; unsupported format produces an error
`SRS-010`	Acquire image from TWAIN device	`SDS-025`	TWAIN image is acquired and displayed correctly in the viewer

5.3 Viewer Functions

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-011`	Display image of a patient on mounted device	`SDS-003`	Main viewer displays the selected patient mount or current image set without misattribution
`SRS-012`	Display images using thumbnail	`SDS-004`	Thumbnail viewer shows available images and selection state correctly

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-013`	Window to select tooth positions to capture	`SDS-005`	Tooth map navigates to the correct image for the selected position
`SRS-014`	Compare selected images	`SDS-006`	Comparison view is presented for the selected images with independent viewing state
`SRS-015`	Move image	`SDS-003`	Image position changes according to pan interaction without losing annotations
`SRS-016`	Zoom image	`SDS-013`	Image zoom level changes in and out as commanded; image remains usable
`SRS-017`	Rotate image	`SDS-014`	Image rotates or flips according to selected option without corrupting display

5.4 Annotation And Measurement

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-018`	Ellipse	`SDS-007`	Ellipse annotation is created, displayed, and retained correctly across viewer sessions
`SRS-019`	Text	`SDS-008`	Text annotation is created, displayed, and retained correctly
`SRS-020`	Arrow	`SDS-009`	Arrow annotation is created, displayed, and retained correctly

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-021`	Pencil	`SDS-010`	Pencil annotation is created, displayed, and retained correctly
`SRS-022`	Angle	`SDS-011`	Angle measurement is created and displayed correctly
`SRS-023`	Length	`SDS-012`	Length measurement is created, calibrated, and displayed correctly

5.5 Output And Exchange

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-024`	Print the selected image	`SDS-015`	Selected image is printed through the standard print path without application error
`SRS-025`	Print the selected image using a DICOM printer	`SDS-016`	Image is transmitted to DICOM printer without application error
`SRS-026`	Export patient image information to file	`SDS-017`	Image export file is created successfully at target location with preserved attribution
`SRS-027`	Export images to media device such as CD or DVD	`SDS-018`	Export to media completes successfully or reports user-visible error
`SRS-028`	Send image to DICOM server through network	`SDS-019`	DICOM transmission completes or failure is detected and reported

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-029`	Import patients from disc	`SDS-001`	Imported patient data is loaded into the local store with validated records

5.6 Device And Workflow Support

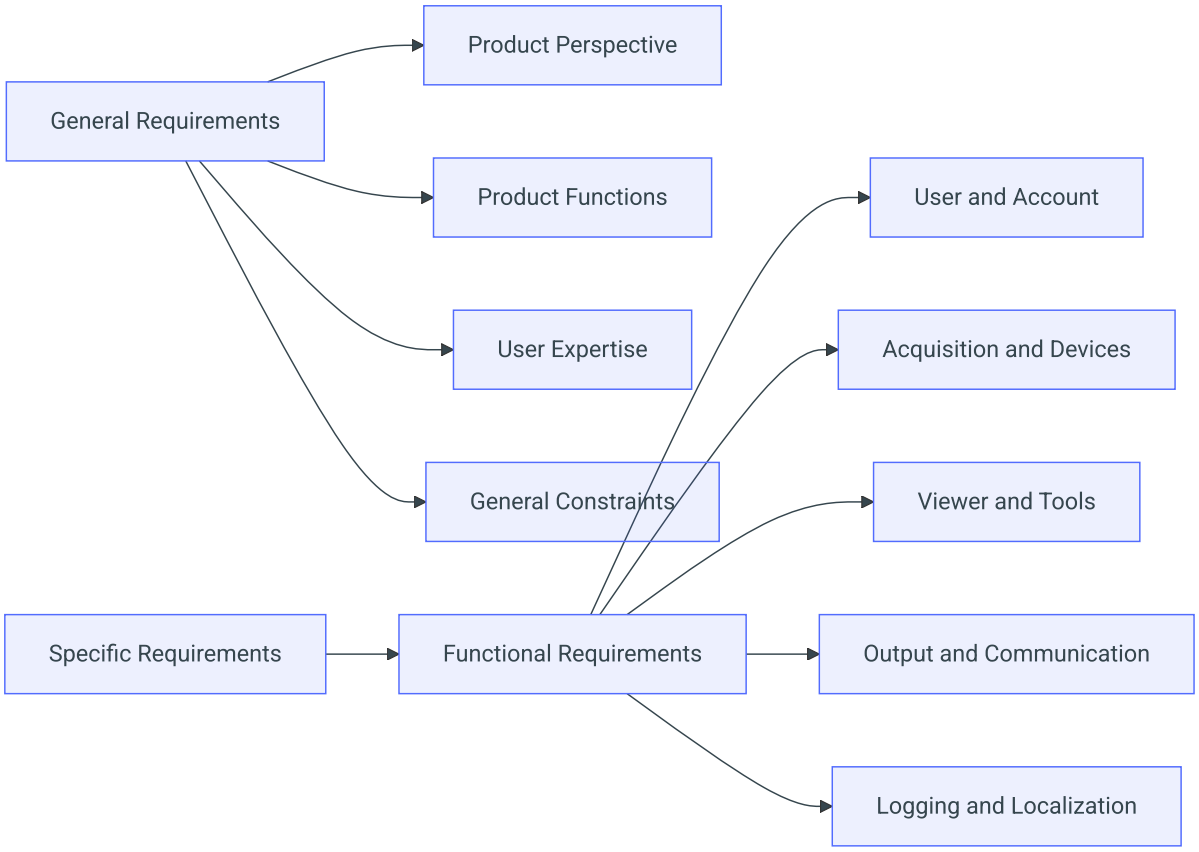
SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-030`	Switch FMX continuous acquisition	`SDS-027`	Continuous-acquisition workflow proceeds according to the selected FMX mode
`SRS-031`	Use multiple sensor	`SDS-028`	Selected sensor becomes active and acquisition path follows the chosen device
`SRS-032`	Display wait status	`SDS-029`	Sensor or device wait status is shown consistently for the current connection state
`SRS-033`	USB data integrity check	`SDS-030`	Integrity-check result is available and unexpected corruption is detected
`SRS-034`	Local network connection to the medical device	`SDS-020`	Network-based connectivity required for device workflow is available and stable

5.7 Logging, Robustness, And Localization

SRS ID	Requirement	Design Allocation	Acceptance Criteria
`SRS-035`	Record activity status	`SDS-021`	Activity log records expected operations without corruption
`SRS-036`	System instability guidelines	`SDS-031`	Error type is classified and troubleshooting guidance is displayed
`SRS-037`	Multilingual support	`SDS-022`	GUI text changes to the selected language without layout or content corruption

6. Requirement Structure

The current software requirement set follows the pattern below.



7. Requirement Coverage Map

The currently visible requirement identifiers can be grouped as follows.

Requirement Group	Representative IDs
User and account handling	`SRS-001`, `SRS-002`, `SRS-003`, `SRS-004`, `SRS-005`
Acquisition and import	`SRS-006`, `SRS-007`, `SRS-008`, `SRS-009`, `SRS-010`
Viewer and display	`SRS-011`, `SRS-012`, `SRS-013`, `SRS-014`, `SRS-015`, `SRS-016`, `SRS-017`
Output and communication	`SRS-024`, `SRS-025`, `SRS-026`, `SRS-027`, `SRS-028`, `SRS-029`
Device support and workflow	`SRS-030`, `SRS-031`, `SRS-032`, `SRS-033`, `SRS-034`
Logging and localization	`SRS-035`, `SRS-036`, `SRS-037`

8. Downstream Use

This software requirement set is intended to drive:

- detailed design allocation in `SwSDS for Portview`
- procedure selection in `SwSTP for Portview` and `SwTP for Portview`
- traceability coverage in the Portview traceability matrix
- verification planning and reporting in `SV-04` and `SV-05`